Targeted v.s. Universal Fiscal Policy

Who should receive transfers? A HANK Approach

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Final Project Presentation

Computational Methods in Economics
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Motivation

Why does fiscal targeting matter?

The importance of Fiscal Transfers

- Fiscal Transfers are important police instruments.
- Standard models assume transfers affect all households equally.
- But households differ dramatically in their MPC, wealth, and unemployment status. (Kaplan & Violante, 2014)

Fiscal Transfer's Literature in HANK

- HANK models show that fiscal multiplier depends on who receives the transfer (Auclert et al., 2024, 2025; Kaplan & Violante, 2014).
- Most studies show that the effects depend on how the police is financed — whether by debt or by taxes (McKay & Reis, 2016).
- But no paper has studied all relevant targeting criteria (wealth, income, employment, HtM status) within the same framework.

What do we know?

Heterogeneous Models

- RANK models are insufficient to understand target polices because they do not explain the mismatch with MPCs' data (Kaplan & Violante, 2014).
- HANK was built for monetary policy, but its fiscal implications are equally important and less explored (Kaplan et al., 2018).
- Debt-financed fiscal police has greater aggregated effect than in RANK.

Fiscal Police Design

- Groups overlap but are not identical: a wealthy HtM is very different from a poor unemployed.
- Without considering the mechanism behind heterogeneity, it is impossible to understand the best fiscal direction.

Research Question

Research Question

Does the optimal fiscal transfer rule depend on how the transfer is financed — or is one targeting design robustly better?

To answer this, we compare welfare and aggregate effects across five targeting rules:

1. wealth-based,
2. income-based,
3. employment-based,
4. HtM-status-based,
5. and universal transfers,

under two financing regimes (debt or tax) and two monetary stances (aggressive or accommodative Taylor rule).

Also, we consider a negative aggregate demand shock that triggers the transfers.

Structural Model

Model Setup: A one-asset HANK model with unemployment.

Households (Aiyagari-style)

- Exogenous employment state: $s \in \{emp,\ unemp\}$;
- Idiosyncratic income risk for employed z;
- Optimization (if employed):

$$\max\ \mathbb{E}_0 \sum_t \beta_i^t\, u(c_{it})$$

$$\text{s.t.} \quad c_{it} + a_{it} \leq (1 + r_t^p) a_{it-1} + (1 - \tau_{it}) w_t z_{it} n_{it}$$

Firm (New Keynesian-style)

- Calvo's sticky wages approach;
- Production based on labor;
- Leads to the Phillips curve.

Government

- Targeted transfers: $T = T(a, z, s)$;

$$G_t + B_t + T_t(a, z, u) = (1 + r_{t-1}) B_{t-1} + \int \tau_{it} w_t n_{it}\, d\Lambda(a, z, s)$$

Structural Estimation

Computational Method

- Solve the model using EGM, discretising asset and states by the Rouwenhorst.
- Dynare solves the model via the sequence-space Jacobian (linearization around steady state) (Auclert et al., 2021; Bayer et al., 2024).

Structural Estimation

- Fix and calibrate for unimportant moments (Auclert et al., 2025).
  - Example: two-point β calibrating by generational turnover (Krusell & Smith, 1998).
- Structural calibration by SMM targeting MPC distribution, wealth-to-income ratio, unemployment rate.
- Weighting Matrix estimated by diagonal form.

Planned Exercises

What is planned to be done:

01 Core Idea

Compare welfare and aggregate effects between designs (target or universal) and financing rules (debt or tax).

02 Monetary Extension

How does the monetary police rule (aggressive or accommodative) change the results?

03 Universal Rule

Within universal transfers, does the distributional schedule matter — are progressive transfers better than lump-sum?

04 Transfer Size

Are high-MPC households the key determinant, or does budget size interact with the targeting rule?

Thank You!

Questions? Comments? Suggestions?

Outline

1 References & Additional Slides

References I

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